

Relay Controller – Serial Commands and Pinout Guide

Overview

This controller firmware supports both **Arduino Pro Mini** and **Arduino Pro Micro**, maintaining a consistent physical pinout across both boards.

Serial Commands

All commands are sent over **Serial** at **9600 baud**. Commands are case-insensitive.

Command List

Command	Arguments	Description	
<code>set-timeout {color} {seconds}</code>	Color name, Timeout in seconds	Set relay auto-off timeout.	
<code>set-debounce {color} {milliseconds}</code>	Color name, Debounce time in ms	Set button debounce time.	
<code>on {color}</code>	Color name	Turn a relay ON manually.	
<code>off {color}</code>	Color name	Turn a relay OFF manually.	
<code>save</code>	None	Save current timeout and debounce settings to EEPROM.	
<code>show-timeout {color}</code>	Color name	Show current timeout value in runtime config.	
<code>show-debounce {color}</code>	Color name	Show current debounce value in runtime config.	
<code>load-timeout {color}</code>	Color name	Load timeout value from EEPROM into runtime config.	
<code>load-debounce {color}</code>	Color name	Load debounce value from EEPROM into runtime config.	
<code>status {color}</code>	Color name	Get current latch status and timer ticks.	
<code>`logging on</code>	<code>off</code>	"on" or "off"	Enable or disable periodic status logging.

Command	Arguments	Description
<code>read</code>	None	Read raw EEPROM data for all colors (debug/diagnostic).
<code>show</code>	None	Show current runtime config for all colors.
<code>load</code>	None	Load all settings from EEPROM into runtime config.
<code>reset</code>	None	Reset runtime config to defaults (does not write to EEPROM).
<code>help</code>	None	Display list of available commands.

Example Usage

```

set-timeout blue 600    // Set blue relay timeout to 10 minutes
set-timeout red 0       // Disable automatic timeout for red relay
set-debounce red 250    // Set red debounce to 250 ms
on green                // Turn green relay ON
off yellow              // Turn yellow relay OFF
save                   // Save current settings to EEPROM
show-timeout blue       // Show blue timeout from runtime config
load-debounce red       // Load red debounce from EEPROM to runtime config
status red             // Show red relay status
read                   // Print raw EEPROM data for diagnostics
show                   // Show all runtime config values
load                   // Load all config from EEPROM
reset                  // Reset runtime config to default values
logging off            // Disable periodic logging
help                   // Show available commands

```

Compatible Boards

- ☒ **Arduino Pro Mini** (5V, ATmega328P)
- ☒ **Arduino Pro Micro** (5V, ATmega32U4)

Note: This firmware automatically adjusts for board differences and maintains a consistent **physical pin mapping** for relay/buttons/LEDs across both boards.

Relay/Button/LED Pin Mapping

Signal	Pro Mini Pin	Pro Micro Pin	Description
Blue Button	A1 (15)	F4 (A1)	Input button for blue relay
Green Button	A0 (14)	F5 (A0)	Input button for green relay
Yellow Button	A2 (16)	F6 (A2)	Input button for yellow relay
Red Button	D10 (10)	B6 (D10)	Input button for red relay
Blue LED	A3	F7	Status LED for blue relay
Green LED	A2	F6	Status LED for green relay
Yellow LED	A1	F5	Status LED for yellow relay
Red LED	A0	F4	Status LED for red relay
Blue Relay Control	D6	D7	Output to control blue relay
Green Relay Control	D7	E6	Output to control green relay
Yellow Relay Control	D8	B4	Output to control yellow relay
Red Relay Control	D9	B5	Output to control red relay

Notes:

- Button pins use `INPUT_PULLUP` mode.
- Relays are **active LOW** (0 = ON, 1 = OFF).
- LEDs are **active HIGH**.

Default Settings

Setting	Value
Timeout	30 minutes (1800 seconds)
Debounce	200 milliseconds

EEPROM is initialized with these values on first run or if corrupted data is detected.

EEPROM Storage Map

Address	Size	Description
0	1 byte	EEPROM Version
1	4 bytes	Blue Relay Timeout
5	4 bytes	Blue Relay Debounce

Address	Size	Description
9	4 bytes	Green Relay Timeout
13	4 bytes	Green Relay Debounce
17	4 bytes	Yellow Relay Timeout
21	4 bytes	Yellow Relay Debounce
25	4 bytes	Red Relay Timeout
29	4 bytes	Red Relay Debounce

Note: EEPROM writes only occur on changes or **save** command, reducing wear.

Serial Logging Output

When logging is enabled, the system outputs status every second:

```
blue> latch:0 tick:0 off
green> latch:1 tick:1234567 on
yellow> latch:0 tick:0 off
red> latch:0 tick:0 off
```

Where:

- **latch**: logic state (1 = on, 0 = off)
- **tick**: time in milliseconds when relay was activated
- Final word: actual relay output state

A built-in LED blinks once per second to indicate the controller is running.